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Regulating ChatGPT and other Large Generative AI Models

PHILIPP HACKER, European University Viadrina of Frankfurt (Oder), Frankfurt/Oder, Brandenburg, Germany

ANDREAS ENGEL, Heidelberg University, Heidelberg, Baden-Wurttemberg, Germany

MARCO MAUER, Humboldt University of Berlin, Berlin, Germany

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Philipp Hacker
European New School of Digital Studies, European University Viadrina
hacker@europa-uni.de

Andreas Engel
Heidelberg University
andreas.engel@igw.uni-heidelberg.de

Marco Mauer
Humboldt University of Berlin,
marco.mauer@hu-berlin.de
with limited input from ChatGPT (see Acknowledgments)

ABSTRACT

Large generative AI models (LGAIMs), such as ChatGPT, GPT-4 or Stable Diffusion, are rapidly transforming the way we communicate, illustrate, and create. However, AI regulation, in the EU and beyond, has primarily focused on conventional AI models, not LGAIMs. This paper will situate these new generative models in the current debate on trustworthy AI regulation, and ask how the law can be tailored to their capabilities. After laying technical foundations, the legal part of the paper proceeds in four steps, covering (1) direct regulation, (2) data protection, (3) content moderation, and (4) policy proposals. It suggests a novel terminology to capture the AI value chain in LGAIM settings by differentiating between LGAIM developers, deployers, professional and non-professional users, as well as recipients of LGAIM output. We tailor regulatory duties to these different actors along the value chain and suggest strategies to ensure that LGAIMs are trustworthy and deployed for the benefit of society at large. Rules in the AI Act and other direct regulation must match the specificities of pre-trained models. The paper argues for three layers of obligations concerning LGAIMs (minimum standards for all LGAIMs; high-risk obligations for high-risk use cases; collaborations along the AI value chain). In general, regulation should focus on concrete high-risk applications, and not the pre-trained model itself, and should include (i) obligations regarding transparency and (ii) risk management. Non-discrimination provisions (iii) may, however, apply to LGAIM developers. Lastly, (iv) the core of the DSA's content moderation rules should be expanded to cover LGAIMs. This includes notice and action mechanisms, and trusted flaggers.

CCS CONCEPTS

• **Social and professional topics** → Computing / technology policy; Government / technology policy; Governmental regulations; • **Additional Keywords and Phrases: LGAIMs, LGAIM regulation, general-purpose AI systems, GPAIS, foundation models, large language models, LLMs, AI regulation, AI Act, direct AI regulation, data protection, GDPR, Digital Services Act, content moderation**;

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1 INTRODUCTION

Large generative AI models (LGAIMs) are rapidly transforming the way we communicate, create, and work. Their consequences are bound to affect all sectors of society, from business development to medicine, from education to research, and from coding to entertainment and the arts. LGAIMs harbor great potential, but also carry significant risk. Today, they are relied upon by millions of users to generate human-level text (e.g., GPT-4, ChatGPT, Luminous, Bard, Bing), images (e.g., Stable Diffusion, DALL·E 2), videos (e.g., Synthesia), or audio (e.g., MusicLM), while further alternatives are already in the pipeline [1-3]. Soon, they may be part of employment tools ranking and replying to job candidates, or of hospital administration systems drafting letters to patients based on case files. Freeing up time for professionals to focus on substantive matters–for example, actual patient treatment–, such multi-modal decision engines may contribute to a more effective, and more just, allocation of resources. However, errors will be costly, and risks ranging from discrimination and privacy to disrespectful content need to be adequately addressed [4-6]. Already now, LGAIMs' unbridled capacities may be harnessed to take manipulation, fake news, and harmful speech to an entirely new level [7-11]. As a result, the debate on how (not) to regulate LGAIMs is becoming increasingly intense [12-22].

In this paper, we argue that regulation, and EU regulation in particular, is not only ill-prepared for the advent of this new generation of AI models, but also sets the wrong focus by quarreling mainly about direct regulation in the AI Act at the expense of the, arguably, more pressing content moderation concerns under the Digital Services Act (DSA). AI regulation, in the EU and beyond, has primarily focused on conventional AI models, not on the new generation whose birth we are witnessing today. The paper will situate these new generative models in the current debate on trustworthy AI regulation, and ask what novel tools might be needed to tailor current and future law to their capabilities. Inter alia, we suggest that the terminology and obligations in the AI Act and other pertaining regulation be further differentiated to better capture the realities of the evolving AI value chain. Some of these observations also apply to traditional AI systems; however, generative models are special in so far as they create output designed for communication or speech–and thus raise important and novel questions concerning the regulation of AI-enabled communication, which we analyze through the lens of the DSA and, in the technical report, non-discrimination law and the GDPR.

To do so, the paper proceeds in five steps. First, we cover technical foundations of LGAIMs, and typical scenarios of their use, to

the extent that they are necessary for the ensuing legal discussion. Second, we critique the EU AI Act, which seeks to directly address risks by AI systems. The versions adopted by the Council (Art. 4a-c AI Act[1]) and the European Parliament (Art. 28-28b AI Act EP Version[2]) contain provisions to explicitly regulate LGAIMs, even if their providers are based outside of the EU [14, cf. also 23]. These proposals, however, arguably fail to fully accommodate the capacities and broad applicability of LGAIMs, particularly concerning the obligation for an encompassing risk management system covering all possible high-risk purposes (Art. 9 AI Act; Art. 28b(1)(a) AI Act EP Version) [12, pp. 6-10, 24, pp. 13, 51 et seqq.]. Third, we briefly touch on non-discrimination and data protection law (more detail in the Technical Report). Fourth, we turn to content moderation [see, e.g., 25, 26, 27]. Recent experiments have shown that ChatGPT, despite innate protections [28], may be harnessed to produce hate speech campaigns at scale, including the code needed for maximum proliferation [8]. Furthermore, the speed and syntactical accuracy of LGAIMs make them the perfect tool for the mass creation of highly polished, seemingly fact-loaded, yet deeply twisted fake news [7, 17]. In combination with the factual dismantling of content moderation on platforms such as Twitter, a perfect storm is gathering for the next global election cycle. We show that the EU's prime instrument to combat harmful speech, the DSA [29, 30], does not apply to LGAIMs, creating a dangerous regulatory loophole.

Finally, the paper argues for three layers of obligations concerning LGAIMs (minimum standards for all LGAIMs; high-risk obligations for high-risk use cases; collaborations along the AI value chain; cf. now also Art. 28 and 28b AI EP Version) and makes four specific policy proposals to ensure that LGAIMs are trustworthy and deployed for the benefit of society at large: direct regulation of LGAIM deployers and users, including (i) transparency and (ii) risk management; (iii) the application of non-discrimination provisions to LGAIM developers; and (iv) specific content moderation rules for LGAIMs. We conclude with a brief assessment concerning the vice and virtue of technology-neutral regulation.

Due to space constraints, we cannot address all social and regulatory concerns regarding LGAIMs and have to bracket, for example, questions of IP law, power dynamics [31, 32], a deeper exploration of the comparative advantages of technology-neutral and technology-specific regulation [33], or the use of LGAIMs in military contexts [34, 35].

2 TECHNICAL FOUNDATIONS OF LARGE GENERATIVE AI MODELS AND EXEMPLARY USAGE SCENARIOS

The AI models covered by this contribution are often referred to as 'foundation models' [36], 'large language models' (LLMs) [37] or 'large generative AI models' (LGAIMs–the term adopted in this article) [38]. Although the emergence of these models in recent years constitutes a significant technical advance (for foundations, see [39-43]), they harness, to a reasonable extent, existing technologies in a vastly increased scale and scope. LGAIMs are usually trained with several billion, if not hundreds of billions, parameters [43, 44], requiring large amounts of training data and computing power [45]. While there are ongoing research efforts to make training language models, and, in particular, transformers, more efficient [46, 47], the energy required to train models this large has triggered concerns from a climate policy perspective [24, 48-52] (see also Part 6).

Hence, LGAIMs "*are advanced machine learning models that are trained to generate new data, such as text, images, or audio*" (Prompt 1, see Annex H1). This "*makes them distinct from other AI models [. . . only] designed to make predictions or classifications*" (Prompt 2) or to fulfil other specific functions. This increased scope of application is one of the reasons for the large amount of data and compute required to train them. LGAIMs employ a variety of techniques [28, 53] that aim at allowing them *"to find patterns and relationships in the data on its own, without being [explicitly] told what to look for.*

Once the model has learned these patterns, it can generate new examples that are similar to the training data" (Prompt 3). In simple terms, training data are represented as probability distributions. By sampling from and mixing them, the model can generate content beyond the training data set–thus something new, as some commentators put it [54, 55]. LGAIMs can often digest human text input [56, 57] and produce an output (text; image; audio; video) based on it. The vast amounts of data required imply that developers of LGAIMs must often rely on training data that is openly available on the internet, which can hardly be considered perfect from a data quality perspective [58]. The content generated by these models can, therefore, be biased, prejudiced, or harmful [15, 59]. To avoid or at least mitigate this issue, model developers need to use proper curating techniques [60, 61]. OpenAI, controversially, hired a large content moderation team in Kenya [62].

"[*L]arge generative models can generate synthetic content that is difficult to distinguish from real content, making it challenging to differentiate between real and fake information. [. . . T]he sheer volume of content generated by these models can make it difficult to manually review and moderate all of the generated content*" (Prompt 4). For as much as we know [28], and according to ChatGPT itself, the creators of ChatGPT sought to address this problem by using "*a combination of techniques to detect and remove inappropriate content. This process includes pre-moderation, where a team of human moderators review and approve content before it is made publicly available. Additionally, ChatGPT uses filtering, which involves using natural language processing and machine learning algorithms to detect and remove offensive or inappropriate content. This is done by training a machine learning model on a dataset of examples of inappropriate content, and then using this model to identify similar content in new inputs*" (Prompt 5). While we cannot perfectly verify these claims due to lack of transparency on OpenAI's side, it seems that ChatGPT relied or relies on humans that train an automatic content moderation system to prevent the output from becoming abusive [62].

Even (idealized) automated and perfect detection of abusive content would only solve half the problem, though. What remains is the danger of creating "fake news" that are hard to spot [17]. Regulation arguably needs to tackle these challenges. To better

[1]Unless otherwise noted, all references to the AI Act are to the general approach adopted by the EU Council on Dec. 6, 2022, available under https://data.consilium.europa.eu/doc/document/ST-14954-2022-INIT/en/pdf; we have been able to incorporate policy developments until May 12, 2023.

[2]DRAFT Compromise Amendments on the Draft Report, Proposal for a regulation of the European Parliament and of the Council, Brando Benifei & Ioan-Dragoş Tudorache (May 9, 2023), https://www.europarl.europa.eu/meetdocs/2014_2019/plmrep/COMMITTEES/CJ40/DV/2023/05-11/ConsolidatedCA_IMCOLIBE_AI_ACT_EN.pdf (= AI Act EP Version).

highlight them, for the discussion that follows, we will consider the following two lead examples: in a business context, one might think of a sportswear manufacturer (e.g., adidas or Nike) that wants to use the potential of a LGAIM specifically for the design of clothing. For this purpose, adidas might use a pre-trained model provided by a developer (e.g., Stability AI), while another entity, the deployer, would fine-tune the model according to adidas' requirements (and possibly host it on a cloud platform). As a second exemplary use case, in a private setting, one could think of a young parent that uses an AI text generator to generate a funny (and suitable) invitation text for her daughter's birthday party. To do so, (s)he might consult Aleph Alpha's Luminous or ChatGPT and ask the chatbot to come up with an appropriate suggestion.

3 DIRECT REGULATION OF THE AI VALUE CHAIN: THE EUROPEAN AI ACT

On May 13, 2022, the French Council presidency circulated an amendment to the draft AI Act, Art. 4a-4c, on what the text calls "general-purpose AI systems" (GPAIS). This novel passage has come to form the nucleus of direct regulation of LGAIMs. It was fiercely contested in the EP [63-65] and will be a key point of debate for the final version of the AI Act. The general approach adopted by the Council on December 6, 2022, defines GPAIS as systems "intended by the provider to perform generally applicable functions such as image and speech recognition, audio and video generation, pattern detection, question answering, translation and others; a general purpose AI system may be used in a plurality of contexts and be integrated in a plurality of other AI systems" (Art. 3(1b) AI Act). Under the Council version, GPAIS are subjected to the high-risk obligations (e.g., Art. 8 to 15 AI Act) if they may be used as high-risk systems or as components thereof (Art. 4b(1)(1) and 4b(2) AI Act).

3.1 Critique of the GPAIS AI Act Rules

The AI Act heroically strives to keep pace with the accelerating dynamics in the AI technology space. However, in our view, the recently introduced rules on GPAIS fail to do justice to the peculiarities of large AI models, and particularly LGAIMs, for three reasons.

3.1.1 Toward a Definition of GPAIS. First, the definition in Art. 3(1b) AI Act is significantly over-inclusive. Rules on GPAIS were inspired by the surge in the release of and literature on foundation models and LGAIMs. As seen in Part 2, LGAIMs operate with large numbers of parameters, training data, and compute. Significantly, they generally operate on a wider range of problems than traditional models do [43]. Conceptually, their "generality" may refer to their *ability* (e.g., language versus vision, or combinations in multimodal models); *domain* of use cases (e.g., educational versus economic); breadth of *tasks* covered (e.g., summarizing versus completing text), or versatility of *output* (e.g., black and white versus multicolored image) [14]. GPAIS, in our view, must necessarily display significant generality in ability, tasks, or outputs, beyond the mere fact that they might be integrated into various use cases (which also holds true for extremely simple algorithms). The broad definition of GPAIS in the AI Act (Council general approach) clashes with this understanding, however. According to that rule, every simple image or speech recognition system seems to qualify, irrespective of the breadth of its capabilities; rightly, this only corresponds to a minority position in the technical GPAIS literature [14, 66].

3.1.2 Risk Management for GPAIS. Second, even a narrower definition would not avoid other problems. Precisely because large AI models are so versatile, providers will generally not be able to avail themselves of the exception in Art. 4c(1) AI Act: by excluding all high-risk uses, they would not act in good faith, as they would have to know that the system, once released, may and likely will be used for at least one high-risk application. For example, language models may be used to summarize or rate medical patient files, or student, job, credit or insurance applications (Annexes II, Section A. No. 12, 13 and III No. 3-5 AI Act. Unless any misuse can be verifiably technically excluded, LGAIMs will therefore generally count as high-risk systems under the proposed provision.

This, however, entails that they have to abide by the high-risk obligations, in particular the establishment of a comprehensive risk management system, according to Art. 9 AI Act. Setting up such a system seems to border on the impossible, given LGAIMs' versatility. It would compel LGAIM providers to identify and analyze all "known and foreseeable risks most likely to occur to health, safety and fundamental rights" concerning all possible high-risk uses of the LGAIM (Art. 9(2)(a), 4b(6) AI Act). On this basis, mitigation strategies for all these risks have to be developed and implemented (Art. 9(2)(d) and (4) AI Act). Providers of LGAIMs such as GPT-4 would, therefore have to analyze the risks for every single, possible application in every single high-risk case contained in Annexes II and III concerning health, safety and all possible fundamental rights.

Similarly, performance, robustness, and cybersecurity tests will have to be conducted concerning all possible high-risk uses (Art. 15(1), 4b(6) AI Act). This seems not only almost prohibitively costly but also hardly feasible. The entire analysis would have to be based on an abstract, hypothetical investigation, and coupled with–again hypothetical–risk mitigation measures that will, in many cases, depend on the concrete deployment, which by definition has not been implemented at the moment of analysis. What is more, many of these possible use cases will, in the end, not even be realized. Hence, such a rule would likely create "much ado about nothing", in other words: a waste of resources.

3.1.3 Adverse Consequences for Competition. Third, the current GPAIS rules would likely have significantly adverse consequences for the competitive environment surrounding LGAIMs. The AI Act definition specifically includes open source developers as LGAIM providers, of which there are several.[3] Some of these will explore LGAIMs not for commercial, but for philanthropic or research reasons. While, according to its Art. 2(7), the AI Act shall not apply to any (scientific, see Recital 12b AI Act) research and development activity regarding AI systems, this research exemption arguably does not apply anymore once the system is released into the wild (cf. Recital 12b AI Act).

As a result, all entities–large or small–developing LGAIMs and placing them on the market will have to comply with the same stringent high-risk obligations, and be subject to the same liability

[3]See, e.g., https://www.kdnuggets.com/2022/09/john-snow-top-open-source-large-language-models.html.

risks under the new product liability framework [24]. Given the difficulty to comply with the AI Act's GPAIS rules, it can be expected that only large, deep-pocketed players (such as Google, Meta, Microsoft/Open AI) may field the costs to release an approximately AI Act-compliant LGAIM. For open source developers and many SMEs, compliance will likely be prohibitively costly. Hence, the AI Act may have the unintended consequence of spurring further anti-competitive concentration in the LGAIM development market. Similar effects have already been established concerning the GDPR [67]. In this sense, the AI Act threatens to undermine the efforts of the Digital Markets Act to infuse workable competition into the core of the digital and platform economy.

3.1.4 Critique of the European Parliament proposal. In the EP, the question of how to regulate large generative AI models significantly delayed the formulation of the EP position on the AI Act. After a lengthy debate, a compromise was reached in late April/early May 2023.[4] The compromise foresees three layers of obligations that apply to generative AI systems [65, 68]. The first layer will apply to the providers (=developers) of a subset of GPAIS denominated "foundation models" (Art. 28b(1)-(3) AI Act EP Version) and generative AI (Article 28b(4) AI Act EP Version). Referring to a well-known term in the computer science community [see, e.g., 36, 69], the EP version defines foundation models as an AI system "that is trained on broad data at scale, is designed for generality of output, and can be adapted to a wide range of distinctive tasks" (Art. 3(1c) AI Act EP Version) [cf. also 36, at 3]. The focus on generality of output and tasks is indeed better suited to capture the specifics of large generative AI models than the vague definition of GPAIS (see Section 3.1.1). In line with suggestions made in this paper, the general obligations for all foundation models include data governance measures, particularly with a view to the mitigation of bias (Art. 28b(2)(b) AI Act EP Version; see Section 4). Furthermore, appropriate levels of performance, interpretability, corrigibility, safety and cybersecurity must be maintained throughout the model's lifecycle. These requirements have to be tested for, documented, and verified by independent experts, Art. 28b(2)(c) AI Act EP Version. Crucially, however, all foundation models also need to implement risk assessments, risk mitigation measures, and risk management strategies with a view to reasonably foreseeable risks to health, safety, fundamental rights, the environment, democracy and the rule of law, again with the involvement of independent experts, Art. 28b(2)(a) AI Act EP Version. Effectively, this requirement is tantamount to classifying foundation models as high-risk per se.

A crucial element of the minimum standards for generative AI is contained in the "ChatGPT Rule" Art. 28b(4) AI Act EP Version. It contains three main elements. (i) The transparency obligation concerning the use of AI is a step in the right direction. It addresses obligations of providers towards users of AI systems. In our view, additionally, obligations of users towards recipients are warranted in some instances to fight the spread of fake news and misinformation (see Section 6.1). (ii) The rule on preventing a breach of EU law, however, arguably does not go far enough. Here, the compliance mechanisms of the DSA should be transferred much more specifically, for example through clear, mandatory notice and action procedures and trusted flaggers (see Section 6.4). (iii) The disclosure of copyrighted material contained in training data may indeed help authors and creators enforce their rights. However, even experts often argue whether certain works are copyrightable at all or not. What must be avoided is that developers who have, e.g., processed 20 million images now have to conduct a full-scale legal due diligence on these 20 million images to decide for themselves whether they are copyrightable or not. Hence, it must therefore be sufficient to disclose, even in an over-inclusive manner, works which *may be* copyrightable, including those for which it is not clear whether they are ultimately copyrightable or not. Otherwise, again, practically prohibitive due diligence costs will arise. The individual author must then decide, when she discovers her work, whether she thinks it is protected by copyright or not.

The second level refers to "new providers" which significantly modify the AI system, Art. 28(1)(b) and (ba) AI Act EP Version. This new provider, which is called deployer in our paper (see Section 3.2.1), assumes the obligations of the former provider upon substantial modification; the new provider takes on this role (Art. 28(1) and (2)(1) AI Act EP Version). A third level of requirements relates to the AI value chain (Art. 28(2)(2) and (2a) AI Act EP Version), in line with suggestions made below in this paper (see Section 3.2.2).

In our view, while containing steps in the right direction, this proposal would be ultimately unconvincing for as it effectively treats foundation models as high-risk applications (cf. Art. 28b(1)(a) and (f) AI Act EP Version). Of course, as noted and discussed in detail below (Part 5), AI output may be misused for harmful speech and acts (as almost any technology). But not only does this seem to be rather the exception than the rule. The argument concerning adverse competitive consequences applies equally here. Under the EP version, risk assessment, mitigation, and management still remain focused on the model itself rather than the use-case specific application (Art. 28b(2)(a) and (f) AI Act EP Version), even though Recital 58a acknowledges that risks related from AI systems can stem from their specific use. Again, this leads to the onerous assessment and mitigation of hypothetical risks that may never materialize–instead of managing risks at the application level where the concrete deployment can be considered.

3.2 Proposal: Focus on Deployers and Users

This critique does not imply, of course, that LGAIMs should not be regulated at all. However, in our view, a different approach is warranted. Scholars have noted that the regulatory focus should shift [12, 13] and move towards LGAIM deployers and users, i.e., those calibrating LGAIMs for and using them in concrete high-risk applications. While some general rules, such as data governance, non-discrimination and cybersecurity provisions, should indeed apply to all foundation models (see Section 4), the bulk of the high-risk obligations of the AI Act should be triggered for specific use cases only and target primarily deployers and professional users.

3.2.1 Terminology: Developers, Deployers, Users, and Recipients. Lilian Edwards, for example, has rightly suggested to differentiate between developers of GPAIS, deployers, and end users [12, see also 24]. In the following, we take this beginning differentiation in the AI value chain one step further. In many scenarios, there will be at least four entities involved, in different roles [cf. 70]. We suggest that the terminology in the AI Act and other pertaining regulation

[4] See note 2.

must be adapted to the evolving AI value chain in the following way.

Developer: this is the entity originally creating and (pre-) training the model. In the AI Act, this entity is called the provider (under some further conditions, see Art. 3(2)). Real-world examples would be OpenAI, Stability, or Google. **Deployer**: this is the entity fine-tuning the model for a specific use case. The AI Act EP Version also uses the term, albeit in a slightly different manner, covering any person or entity using an AI system under its authority, except where the AI system is used in the course of a personal non-professional activity (Art. 3(4) AI Act EP Version); for the purposes of the AI Act EP Version, a deployer can be a (new) provider, Art. 28(2)(1). Note that there could be several deployers (working jointly or consecutively), leading to a true AI value chain similar to OEM value chains. Alternatively, the developer could simultaneously act as a deployer (vertical integration)–just as for the purposes of the AI Act EP Version, a deployer can be a (new) provider, Art. 28(2)(1). **User**: this is the entity actually generating output from an LGAIM, e.g. via prompts, and putting it to use. The user may harness the output in a professional or a non-professional capacity [71, 72]. Potential real-world examples of professional users would be the clothing and sportswear manufacturer from the first lead example, or any other entity from the groups of professional users just listed. Note that any individual making professionally motivated comments online would also count as a professional user in this respect. Finally, some exceptions from the EU consumer definition are in order: for example, employees[5] (and students, for that matter) should presumptively count as professional users when applying LGAIMs for job- or education-related tasks. Particularly concerning negative externalities of AI output, it should not matter whether users are pursuing a dependent or independent professional activity (e.g., Art. 29 AI Act). By contrast, the AI Act largely exempts non-professional users (cf. Art. 2(8) AI Act; the AI Act EP Version contains no general exemption, but excludes non-professionals from the definition of deployers, Art. 3(4)). The parent from the lead example using ChatGPT for birthday party would fall into this category. **Recipient**: this is the entity consuming the product offered by the user.

With this terminology in place, regulatory obligations can be allocated to different types of actors in more nuanced ways. While developers should, to a certain extent, be subject to non-discrimination law and certain data governance provisions (Section 4), we suggest that the focus of regulatory duties should lie on deployers and users, for example concerning risk management systems (Art. 9 AI Act) or performance and robustness thresholds (Art. 15 AI Act) (see also below, Part 6).

3.2.2 The AI Value Chain. Such a shift of the regulatory focus on deployers and users, however, entails several follow-up problems that need to be addressed [12]. First, deployers and users may be much smaller and less technologically sophisticated than LGAIM developers. This is not a sufficient reason to exempt them from regulation and liability, but it points to the importance of designing a feasible allocation of responsibilities along the AI value chain. Recent proposals discussed in the EP point in this direction as well (see Section 3.1.4). Obligations must be structured in such a way that deployers and users can reasonably be expected to comply with them, both by implementing the necessary technological adjustments and by absorbing the compliance costs. Second, many of the AI Act's high-risk obligations refer to the training and modeling phase conducted, at least partially, by the LGAIM developers. Typically, LGAIM developers will pre-train a large model, which may then be fine-tuned by deployers, potentially in collaboration with developers [73, 74], while users ultimately make the decision what the AI system is used for specifically (e.g. commercial use for design or private use for generating an invitation text). To meet the AI Act requirements concerning training data (Art. 10), documentation and record-keeping (Art. 11 and 12), transparency and human oversight (Art. 13 and 14), performance, robustness and cybersecurity (Art. 15), and to establish the comprehensive risk management system (Art. 9), any person responsible will need to have access to the developer's and deployer's data and expertise. This unveils a regulatory dilemma: focusing exclusively on developers entails potentially excessive and inefficient compliance obligations; focusing on deployers and users risks burdening those who cannot comply due to limited insight or resources. Third, and related to the first and second aspect, individual actors in the AI value chain may simply not have the all-encompassing knowledge and control that would be required if they were the sole addressees of regulatory duties [75]. This more abstract observation also shows that shared and overlapping responsibilities may be needed.

In our view, the only way forward are legally mandated collaborations between LGAIM providers, deployers and users with respect to the fulfillment of regulatory duties. More specifically, we suggest a combination of strategies known from pre-trial discovery, trade secrets law, and the GDPR. Under the current AI Act (Council general approach), such teamwork is encouraged in Art. 4b(5): providers "shall" cooperate with and provide necessary information to users. A key issue, also mentioned in the Article, is access to information potentially protected as trade secrets or intellectual property (IP) rights [13, 76]. To be workable, this obligation needs further concretization; the same holds true for more recent proposals by the EP in this direction [77]; Art. 10(6a) AI Act EP Version only explicitly addresses a situation where such cooperation does not take place.

The problem of balancing collaboration and disclosure with the protection of information is not limited to the AI Act. In our view, it has an internal and external dimension. Internally, i.e., in the relationship between the party requesting and the party granting access, access rights are often countered, by the granting party, by reference to supposedly unsurmountable trade secrets or IP rights [78-80]. The liability directives proposed by the EU Commission, for example, contain elaborate evidence disclosure rules pitting the compensation interests of injured persons against the secrecy interests of AI developers and deployers [24, 81, 82]. Extensive literature and practical experience concerning this problem exists in the realm of the US pretrial discovery system [83-87]. Under this mechanism, partially adopted by the proposed EU evidence disclosure rules [24], injured persons may seek access to documents and information held by the potential defendant before even launching litigation. This, in turn, may lead to non-meritorious access requests by competitors. Similarly, in the AI value chain, developers, deployers and users may indeed not only be business partners but also be (potential)

[5]But see German Constitutional Court, Order of November 23, 2006, Case 1 BvR 1909/06: employees are consumers in the sense of the EU consumer law.

competitors. Hence, deployers' and users' access must be limited. Conversely, some flow of information must be rendered possible to operationalize compliance with high-risk obligations by deployers.

To guard against abuse, we suggest a range of measures. It may be worthwhile to introduce provisions inspired by the US pretrial discovery system [80, 83, 88] and the proposed EU evidence disclosure mechanism (Art. 3(4) AI Liability Directive, protective order). Hence, courts should be empowered to issue *protective orders*, which endow nondisclosure agreements with further weight and subject them to potential administrative penalties. The order may also exempt certain trade secrets from disclosure or allow access only under certain conditions (see F.R.C.P. Rule 26(c)(1)(G)). Furthermore, the appointment of a *special master* may, ultimately, strike a balance between information access and the undue appropriation of competitive advantage (cf. F.R.C.P. Rule 53(a)) [88]. With these safeguards in place, LGAIM developers should be compelled, and not merely encouraged, to cooperate with deployers and users concerning AI Act compliance if they have authorized the deployment.

Concerning the external dimension, the question arises of who should be responsible for fulfilling pertinent duties and be ultimately liable, regarding administrative fines and civil damages, if high-risk rules are violated. Here, we may draw inspiration from Art. 26 GDPR (see also [12]): this mechanism could, mutatis mutandis, be transferred to the AI value chain. Collaboration should be documented in writing to facilitate ex post accountability. Disclosing the core parts of the document, sparing trade secrets, should help potential plaintiffs choosing the right party for following disclosure of evidence requests under the AI liability regime. Finally, joint and several liability ensures collaboration and serves the compensation interests of injured persons. Internally, parties held liable by injured persons can then turn around and seek reimbursement from others in the AI value chain. For example, if the developers essentially retain control via an API distribution model, the internal liability burden will often fall on them. Developers' and deployers' liability, however, must end where their influence over the deployed model ends. Beyond this point, only the users should be the subject of regulation and civil liability (and vice versa, for example in control-via-API cases): incentives for action only make sense where the person incentivized is actually in a position to act [89, 90]. In the GDPR setting, this was effectively decided by the CJEU in the Fashion ID case (CJEU, C-40/17, para. 85). The sole responsibility of the users for certain areas should then also be included in the disclosed agreement to inform potential plaintiffs and foreclose non-meritorious claims against the developer and deployer. Such a system, in our view, would strike an adequate balance of interests and power between LGAIM developers, deployers, users, and affected persons.

The EP version of the AI Act now rightly contains rules on the AI value chain [68]. However, these need to be rendered more specific, as laid out in the preceding sections, to function effectively. Ultimately, allocating responsibility and liability along the value chain is crucial if the AI Act seeks to maintain its spirit of a technology-specific instrument that does not, however, regulate models per se, but primarily models in concrete use cases.

4 NON-DISCRIMINATION LAW AND THE GDPR

Some rules will have to apply directly to LGAIMs and LGAIM developers, however (see Section 6). A clear candidate for such rules is non-discrimination law. Generally, it applies, in the US as well as the EU, in a technology-neutral way [91-95]. Importantly, however, it only covers certain enumerated areas of activity, such as employment, education, or publicly available offers of goods and services [91, 96]. This begs the question whether general-purpose systems may be affected by non-discrimination provisions even before they have been deployed in specific use cases. While a detailed discussion transcends the scope of this paper (see Technical Report), it seems convincing to consider adequate non-discrimination rules a crucial element of any future regulatory perimeter for LGAIMs (Section 6.3).

A third major challenge for any AI model is GDPR compliance. Its relevance for LGAIMs in particular was illustrated by the temporary limitation on the processing of Italian users' data in April 2023. Overall, this measure by the Italian Data Protection Authority rightly points to the legitimate interests, and rights, of data subjects to be informed about how their personal data is used in training and fine-tuning generative AI models (for a more detailed discussion, see the technical report). It should be taken as a welcome wake-up call to the community of developers to share crucial information–on training, personal data, and pertinent risks–with the general public, instead of guarding secrets under the misnomer of OpenAI et al.

5 GENERATIVE MODEL CONTENT MODERATION: THE EUROPEAN DIGITAL SERVICES ACT

The fourth large regulatory frontier concerning LGAIMs is content moderation. Generative models, as virtually any novel technology, may be used for better (think: birthday cards) or worse purposes (think: shitstorm) [97]. The developers of ChatGPT, specifically, anticipated the potential for abuse and trained an internal AI moderator, with controversial help from Kenyan contractors [62] , to detect and block harmful content [12]. AI Research has made progress in this area recently [98-100]. OpenAI has released a content filtering mechanism which users may apply to analyze and flag potentially problematic content along several categories (violence; hate; sexual content etc.).[6] Other large generative models have similar functionalities. However, actors intent on using ChatGPT, and other models, to generate fake or harmful content will find ways to prompt them to do just that. Prompt engineering is becoming a new art to elicit any content from LGAIMs [101] and fake news is harder to detect than hate speech, even though industry efforts are underway via increased model and source transparency [102]. As could be expected, DIY instructions for circumventing content filters are already populating YouTube and reddit,[7] and researchers have already generated an entire hate-filled shitstorm, along with code for proliferation, using ChatGPT [8].

[6] See https://platform.openai.com/docs/api-reference/moderations.
[7] See, e.g., https://www.youtube.com/watch?v=qpKlnYLtPjc; https://www.reddit.com/r/OpenAI/comments/zjyrvw/a_tutorial_on_how_to_use_chatgpt_to_make_any/.

To stem the tide of fake news and hateful content–, the EU has recently enacted the DSA. However, LGAIMs were not in the focus of public attention at the time when the DSA was being drafted. Hence, the DSA was designed to mitigate illegal content on social networks, built by human actors or the occasional Twitter bot, not to counter LGAIMs. The problem lies not in the territorial applicability of the provision: the DSA, like the AI Act, covers services offered to users in the EU, irrespective of where the providers have their place of establishment (Art. 2(1), 3 (d) and (e) DSA).

Yet, the DSA seems outdated at the moment of its enactment due to two crucial limitations in its scope of application. First, it covers only so-called intermediary services (Art. 2(1) and (2) DSA). Art. 3(g) DSA defines them as "mere conduit" (e.g., Internet access providers), "caching" or "hosting" services (e.g., social media platforms, see also Recital 28 DSA). Arguably, however, LGAIMs do not fall into any of these categories. Clearly, they are not comparable to access or caching service providers, which power Internet connections. Hosting services, in turn, are defined as providers storing information provided by, and at the request of, a user (Art. 3(g)(iii) DSA) [see also 103].While users do request information from LGAIMs via prompts, they can hardly be said to provide this information. Rather, other than in traditional social media constellations, it is the LGAIM, not the user, who produces the text. To the contrary, CJEU jurisprudence shows that even platforms merely storing user-generated content may easily lose their status as hosting providers, and concomitant liability privileges under the DSA (and its predecessor in this respect, the E-Commerce Directive), if they "provide assistance" and thus leave their "neutral position", which may even mean merely promoting user-generated content (CJEU, Case C-324/09, L'Oréal para 116). A fortiori, systems generating the content themselves cannot reasonably be qualified as hosting service providers. Hence, the DSA does not apply.

This does not imply that LGAIM content generation is not covered by content liability laws. Rather, its output may be covered by speech regulation, similar to comments made by human users online. However, this branch of the law is largely left to Member State tort law, with the exception of Art. 82 GDPR in the case of processing personal data of victims, which seems rather far-fetched in LGAIM constellations. Not only does such direct speech regulation vary considerably between Member States [104], it also often lacks precisely the instruments the DSA has introduced to facilitate the rapid yet procedurally adequate removal of harmful speech and fake news from the online world: notice and action mechanisms flanked by procedural safeguards; trusted flaggers; obligatory dispute resolution; and comprehensive compliance and risk management regimes for large platforms.

The risk of a regulatory loophole might be partially closed, one might object, by the applicability of the DSA to LGAIM-generated posts that human users, or bots, publish on social networks. Here, the DSA generally applies, as Twitter et al. qualify as hosting service providers. However, a second important gap looms: Recital 14 DSA specifies that the main part of the regulation does not cover "private messaging services." While the notice and action mechanism applies to all hosting services, instruments like trusted flaggers, obligatory dispute resolution, and risk management systems are reserved for the narrower group of "online platforms" [105]. To qualify, these entities must disseminate information to the public (Art. 3(g)(iii), (k) DSA). According to Recital 14 DSA, closed groups on WhatsApp and Telegram, on which problematic content particularly proliferates, are explicitly excluded from the DSA's online platform regulation (Art. 19 et seqq. DSA) as messages are not distributed to the general public. With the right lines of codes, potentially supplied by an LGAIM as well [8], malicious actors posting content in such groups may therefore fully escape the ambit, and the enforcement tools, of the DSA.

Hence, the only action to which the full range of the DSA mechanisms continues to apply is the posting of LGAIM-generated content on traditional social networks. However, at this point in time, Pandora's box has already been opened. Misinformation may also be spread effectively and widely via interpersonal communication. Even if the EU legislator has decided to exclude closed groups from the scope of the DSA [106], this balance needs to be reassessed in the context of readily available LGAIM output, which exacerbates risks. Even the most stringent application of DSA enforcement mechanisms, potentially coupled with GDPR provisions on erasure of data (Art. 17(2) and 19 GDPR), cannot undo the harm done, and often cannot prevent the forward replication of problematic content [107]. Overall, current EU law, despite the laudable efforts in the DSA to mitigate the proliferation of fake news and hate speech, fails to adequately address the dark side of LGAIMs.

6 POLICY PROPOSALS

The preceding discussion has shown that regulation of LGAIMs is necessary, but must be better tailored to the concrete risks they entail. Hence, we suggest a shift away from the wholesale AI Act regulation envisioned in the general approach of the Council of EU toward specific regulatory duties and content moderation. Importantly, regulatory compliance must be feasible for LGAIM developers large and small to avoid a winner-takes-all scenario and further market concentration [67]. This is crucial not only for innovation and or consumer welfare [29, 108, 109], but also for environmental sustainability. While the carbon footprint of IT and AI is significant and steadily rising [48-52], and training of LGAIMs is particularly resource intensive [110], large models may ultimately create fewer greenhouse gas emissions than their smaller brethren if they can be adapted to multiple uses.

Against this background, we envision three layers of obligations for LGAIMs: the first set of minimum standards for all LGAIMs; a second set of specific high-risk rules applying only to LGAIMs used in concrete high-risk use cases; and the third set of rules governing collaboration along the AI value chain (see Section 3.2.2) to enable effective compliance with the first two sets of rules.

Concerning minimum standards, first and foremost, the EU acquis applies to developers of LGAIMs as well, putting the GDPR and non-discrimination law (Section 4 and Technical Report), as well as product liability [24], center stage. In addition, transparency rules, now also proposed by the EP [65], must apply (see below, Section 6.1). Furthermore, specific risks of such outstanding relevance that they should be addressed at the upstream level, rather than delegated to deployers in specific use cases, must be allocated to developers as part of the minimum standards. This concerns, in our view, selected data governance duties (Art. 10 AI Act, see Section 4) and rules on the ever more important issue of cybersecurity

(Art. 15 AI Act). Finally, sustainability rules [24] as well as content moderation (see below, Section 6.4) should also form part of the minimum standards applicable to all LGAIMs.

In the following, we make four concrete, workable suggestions for LGAIM regulation on the first and second level: (i) transparency obligations (first and second level); (ii) mandatory yet limited risk management (second level); (iii) non-discrimination data audits; and (iv) expanded content moderation.

6.1 Transparency

The AI Act contains a wide range of disclosure obligations (Art. 11, Annex IV AI Act) that apply, however, only to high-risk systems. In our view, given the vast potential and growing relevance of LGAIMs for many sectors of society, LGAIMs should — irrespective of their categorization as high-risk or non-high-risk — be subject to two distinct transparency duties.

6.1.1 Transparency requirements for developers and deployers. First, *LGAIM developers and deployers* should be required to report on the provenance and curation of the training data, the model's performance metrics, and any incidents and mitigation strategies concerning harmful content. Ideally, to the extent technically feasible [48, p. 28, Annex A], they should also disclose the model's greenhouse gas (GHG) emissions, to allow for comparison and analysis by regulatory agencies, watchdog organizations, and other interested parties. This information could also serve as the basis for an AI Sustainability Impact Assessment [24, p. 65 f., see also 111].

6.1.2 Transparency requirements for users. Second, *professional users* should be obligated to disclose which parts of their publicly available content were generated by LGAIMs, or adapted based on their output. Specifically, this entails that in adidas example, adidas needs to adequately inform users that the design was generated using, e.g., Stable Diffusion. While the added value of such information may be limited in sales cases, such information is arguably crucial in any cases involving content in the realm of journalism, academic research, or education. Here, the recipients will benefit from insight into generation pipeline. They may use such a disclosure as a warning signal and engage in additional fact checking or to at least take the content *cum grano salis*. Eventually, we imagine differentiating between specific use cases in which AI output transparency vis-à-vis recipients is warranted (e.g., journalism, academic research or education) and others where, based on further analysis and market scrutiny, such disclosures may not be warranted (certain sales, production and B2B scenarios, for example). For the time being, however, we would advocate a general disclosure obligation for professional users to generate further information and insight into the reception of such disclosures by other market participants or recipients.

Conversely, we submit that *non-professional users* should not be required to inform about the use of AI. In the birthday example, hence, a parent would not need to inform the parents that the invitation or the entire design of the birthday party was rendered possible by, e.g., Aleph Alpha's Luminous or ChatGPT. One might push back against this in cases involving the private use of social media, particularly harmful content generated with the help of LGAIMs. However, any rule to disclose AI-generated content would likely be disregarded by malicious actors seeking to post harmful content. Eventually, however, one might consider including social media scenarios into the domain of application of the transparency rule if AI detection tools are sufficiently reliable. In these cases, malicious posts could be uncovered, and actors would face not only the traditional civil and criminal charges, but additionally AI Act enforcement, which could be financially significant (administrative fines) and hence create even greater incentives to comply with the transparency rule, or refrain from harmful content propagation.

The enforcement of any user-focused transparency rule being arduous, it must be supported by technical measures such as digital rights management and watermarks imprinted by the model [112]. The EP is currently pondering a watermark obligation for generative AI [111]. Importantly, more interdisciplinary research is necessary to develop markings that are easy to use and recognize, but hard to remove by average users [113]. This should be coupled with research on AI-content detection to highlight such output where watermarks fail [99, 114].[8]

6.2 Risk Management and Staged Release

As mentioned, one major obstacle to the effective application of the AI Act to LGAIMs proper is comprehensive risk management. Here, novel approaches are needed. Scholars have rightly suggested that powerful models should be released consciously, trading off the added benefit of public scrutiny with the added risk of misuse in the case of full public releases [69, 115]. Additional factors, such as the balance of power among developers, must also be considered [115]. In our view, a limited, staged release, coupled with only access for security researchers and selected stakeholders, may often be preferable [see also 9, 69, 116, 117]. This adds a nuanced, community-based risk management strategy by way of codes of conduct to the regulatory mix [cf. also 117]. Regulatory oversight could be added by way of "regulated self-regulation;" an approach with potentially binding effect of the code of conduct, à la Art. 40 GDPR, seems preferable to the purely voluntary strategy envisioned in Art. 69 AI Act.

Importantly, the full extent of the high-risk section of the AI Act, including formal risk management, should only apply if and when a particular LGAIM (or GPAIS) is indeed used for high-risk purposes (see Part 3.2). This strategy aligns with a general principle of product safety law [13]: not every screw and bolt must be manufactured to the highest standards. For example, only if they are used for spaceships, stringent product safety regulations for producing aeronautics material apply[9] –but not if they are sold in the local DIY store for generic use. The same principle should be applied to LGAIMs.

6.3 Non-Discrimination and training data

Furthermore, we suggest that, as an exception to the focus on LGAIM deployers, certain data curation duties, for example representativeness and approximate balance between protected groups (cf. Art. 10 AI Act), should apply to LGAIM developers. Discrimination, arguably, is too important a risk to be delegated to the user

[8] See also https://openai.com/blog/new-ai-classifier-for-indicating-ai-written-text/.
[9] See, e.g., product standards, aerospace series, DIN EN 4845–4851 (December 2022) on screws.

stage and must be tackled during development and deployment. Wherever possible, discrimination AI systems should be addressed at its roots (often the training data) and not propagated down the ML pipeline or AI value chain. After all, discriminatory output should, in our view, be avoided in all use cases, even on birthday cards. The regulatory burden, however, must be adapted to the abstract risk level and the compliance capacities (i.e., typically the size) of the company. For example, LGAIM developers should have to pro-actively audit the training data set for misrepresentations of protected groups, in ways proportionate to their size and the type of training material (curated data vs. Twitter feeds scraped from the Internet), and implement feasible mitigation measures. At the very least, real-world training data ought to be complemented with synthetic data to balance historical and societal biases contained in online sources. For example, content concerning professions historically reserved for one gender (nurse; doctor) could be automatically copied and any female first names or images exchanged by male ones, and vice versa, creating a training corpus with more gender-neutral professions for text and image generation.

6.4 Content Moderation

One of the biggest challenges for LGAIMs is, arguably, their potential misuse for disinformation, manipulation, and harmful speech. In our view, the DSA rules conceived for traditional social networks must be expanded and adapted accordingly.

6.4.1 Selective expansion of the DSA to LGAIMs. The EP has partially addressed this challenge by stipulating that foundation models must not violate EU law [76]. In our view, however, regulation should go one step further by selectively expanding DSA rules to LGAIM developers and deployers. LGAIMs, and society, would benefit from mandatory notice and action mechanisms, trusted flaggers, and comprehensive audits for models with particularly many users. The regulatory loophole is particularly virulent for LGAIMs offered as standalone software, as is currently the case. In the future, one may expect an increasing integration into platforms of various kinds, such as search engines or social networks, as evidenced by LGAIM development or acquisition by Microsoft, Meta, or Google. While the DSA would then technically apply, it would still have to be updated to ensure that LGAIM-generated content is covered just like user-generated content. In particular, as LGAIM output currently is particularly susceptible to being used for the spread of misinformation, it seems advisable to require LGAIM-generated content to be flagged as such–if technically feasible. Doctrinally, this could be achieved via an amendment of the DSA or of Art. 29 AI Act, which already contains notification duties in its para. 4 (see Part. 4). Given the current political process in the EU, the latter option seems more realistic.

6.4.2 Implementation in practice. How could DSA-style content moderation applied to ChatGPT et al. look like in practice? We envision it to have two components. These components would combine centralized and decentralized monitoring within a notice-and-action mechanism (cf. Art. 16 DSA).

The first component harnesses the wisdom of the crowd, as it were, to correct LGAIM output. Users should be enabled to flag problematic content and give notice. A special status should be given to a specific group of users, trusted flaggers (cf. Art. 22 DSA), who could be private individuals, technologies savvy NGOs, or volunteer coders. After registering with the competent authority, they would essentially function as a decentralized content monitoring team. They could experiment with different prompts and see if they manage to generate harmful or otherwise problematic content. They could also scan the internet for tools to circumvent content moderation policies and instruments at LGAIMs.[10] If they find something, trusted flaggers would send a notice containing the prompt and the output to a content moderation check-in point of the respective LGAIM system, which would forward the notice to developers and/or deployers.

Here, the second component enters the scene, geared toward tech engineers working with developers or deployers. They would have to respond to notices; those submitted by trusted flaggers would have to be prioritized by the content moderation team. Their job, essentially, is to modify the AI system, or to block its output, so that the flagged prompt does not generate problematic output anymore, and to generally search for ways to block easy workarounds likely tried by malicious actors. Furthermore, if the LGAIM system is large enough, they would be tasked with establishing a more comprehensive compliance system (cf. Art. 34-35 DSA). Overall, such a combination of centralized and decentralized monitoring could prove more effective and efficient than current systems relying essentially on goodwill to handle the expected flood of hate speech, fake news and other problematic content generated by LGAIMs.

6.5 Outlook: Technology-specific vs. technology-neutral regulation

Overall, we have added several policy proposals. As a matter of regulatory technique, the legislator should, in our view, strive to shift its strategy from technology-specific regulation–which will often be outdated before eventually enacted–toward more technology-neutral regulation wherever possible. Due to space constraints, we cannot elaborate on this point. However, future analysis may show that non-discrimination law, formulated in a technology-neutral way, continues to grapple with various challenges, but arguably does a better job capturing the dynamics of LGAIM development than the AI Act or the DSA, at least in the way they are currently enacted and proposed (see also the Technical Report). While technology-neutral regulation must be tailored, via agency decisions, regulatory guidelines, and court judgments, to specific technologies, such "small-scale" adaptations are, arguably, often faster to produce than changes to a formal, technology-specific, legislative act. For example, to extend the DSA to LGAIMs in specific ways, one would have to update the DSA or include a reference in the AI Act. Both modifications require concurring decisions by the EP and the Council (Art. 289 TFEU). In non-discrimination law, by contrast, all that is needed, in principle, is an adequate interpretation of existing law by agencies and courts. Their decisions, at least in lower courts, can potentially be rendered faster and be used more flexibly to carve out (preliminary[11]) safe harbors for developers,

[10] See Fn. 7.

[11] Ultimately, we agree that it may take a substantial amount of time for final decisions to emerge from the court system. Only these can deliver a higher degree of legal certainty. However, even lower-court judgments or agency decisions may, arguably,

deployers, and users, and to establish red lines to protect affected persons.

7 CONCLUSION

Scholars and regulators have long suggested that technology-neutral laws may be better prepared to tackle emerging risks given the rapid pace of innovation in machine learning [118-120]. While this claim, arguably, cannot be generally affirmed or refuted, LGAIMs offer a cautionary example for regulation focused specifically on certain technologies. As our study shows, technology-neutral laws sometimes fare better because technology-specific regulation (on platforms; AI systems) may be outdated before (AI Act, AI liability regime) or at the moment of its enactment (DSA). Overall, we add several policy proposals to the emerging regulatory landscape surrounding LGAIMs.

To start with, we argue for a new, differentiated terminology to capture the relevant actors in the AI value chain, in LGAIM settings and beyond. These include: LGAIM developers, deployers, professional and non-professional users, as well as recipients of LGAIM output. Such a nuanced understanding is necessary to allocate regulatory duties to specific actors and activities in the AI value chain. The general approach adopted by the Council of the EU failed to address the specificities of the LGAIM value chain. Rules in the AI Act and other direct regulation must match the specificities of pre-trained models.

More concretely, we propose three layers of rules applicable to LGAIMs. The first layer applies directly to all LGAIMs. It comprises existing, technology neutral regulation such as the GDPR or non-discrimination provisions. Arguably, a version of Art. 10 AI Act and of the cybersecurity rules in Art. 15 AI Act should also apply to LGAIM developers. Furthermore, sustainability and content moderation instruments also form part of this first layer. Art. 28b AI Act EP Version represents an imperfect step into this direction.

On the second layer, we suggest generally singling out concrete high-risk applications, and not the pre-trained model itself, as the object of high-risk obligations. For example, it seems inefficient and practically infeasible to compel the developers of ChatGPT et al. to draw up a comprehensive risk management system covering, and mitigating, all the hypothetical risks to health, safety and fundamental rights such LGAIMs may pose – as the AI Act EP Version still does (Art. 28b(1)(a) and (f)). Rather, if used for a concrete high-risk purpose (e.g., summarizing or grading résumés in employment decisions), the specific deployer and user should have to comply with the AI Act's high-risk obligations, including the risk management system.

The devil, however, is in the detail: providers need to cooperate with deployers to comply with even such narrower regulatory requirements. Hence, a third layer mandating collaboration between actors in the AI value chain for compliance purposes is necessary. Here, we suggest drawing on experience from the US pretrial discovery system and Art. 26 GDPR to balance interests in the access to information with trade secret protection. Art. 28(2a) AI Act EP Version has partly taken up this proposal.

The last section makes concrete policy proposals. For example, detailed transparency obligations are warranted. This concerns both LGAIM developers and deployers (performance metrics; harmful speech issues arisen during pre-training) as well as users (disclosure of the use of LGAIM-generated content).

Finally, the core of the DSA's content moderation rules should be expanded to cover LGAIMs. Art. 28b(4)(b) and generative AI (Article 28b(4) AI Act EP Version) moves in this direction. More specifically, however, rules must also include notice and action mechanisms, trusted flaggers, and, for very large LGAIM developers, comprehensive risk management systems and audits concerning content regulation. Arguably, it is insufficient to tackle AI-generated hate speech and fake news ex post, once they are posted to social media. At this point, their effect will be difficult to stop. Rather, AI generation itself must be moderated by an adequate combination of AI tools, developer and user interventions, and law.

In all areas, regulators and lawmakers need to act fast to keep track with the unchained dynamics of GPT-4 et al. Updating regulation is necessary both to maintain the civility of online discourses and to create a level playing field for developing and deploying the next generation of AI models, in the EU and beyond.

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Passages taken over from ChatGPT are found in the section on technical foundations, all italicized and marked with "", and referenced by the prompt used. They were all collected on January 17, 2023. We deem them factually correct unless otherwise noted. This paper benefitted from comments by Johannes Otterbach, Sandra Wachter, and audiences at AI Campus Berlin, Bucerius Law School (Hamburg), Magdalen College (Oxford), University of Hamburg, and Weizenbaum Institute of the Connected Society. All errors remain entirely our own.

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indicate useful directions and, at least, be used to model compliance tools accordingly, even if policies may have to be revised if decisions are reversed in higher instances.

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PROMPTS

Prompt 1: What are large generative AI models?

Prompt 2: What distinguishes large generative AI models from other AI systems?

Prompt 3: Can you explain the technical foundations of large generative models in simple terms, so that an inexperienced reader understands it?

Prompt 4: What are the objectives, what are the obstacles when it comes to content moderation within large generative AI models?

Prompt 5: How does content moderation work at ChatGPT?